

Mairéad E. Heiger

POSTDOCTORAL FELLOW · DEPARTMENT OF ASTRONOMY, UNIVERSITY OF VIRGINIA

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Education and Research Employment

University of Virginia

COSMICAI AND GECO (GALAXY EVOLUTION AND COSMOLOGY INITIATIVE) FELLOW

- Supervisor: Nitya Kallivayalil

Charlottesville, VA

2026 - Present

University of Toronto

PHD IN ASTRONOMY & ASTROPHYSICS

- Advisors: Ting S. Li, Joshua S. Speagle

Toronto, ON, Canada

2021 - 2026

University of Pittsburgh

BS IN PHYSICS; MINORS IN MATHEMATICS AND ECONOMICS

- Honors: summa cum laude, Phi Beta Kappa, Sigma Pi Sigma

Pittsburgh, PA

2016 - 2020

Publications

PUBLISHED

M. Heiger et al., “It’s a matter of time: Empirical Constraints on Supernova Yields and Delay Times from Dwarf Spheroidal Galaxies”, *ApJ*, vol 1007, no. 2, id. 139, **2026**. [doi:10.3847/1538-4357/ae88f8](https://doi.org/10.3847/1538-4357/ae88f8).

M. Heiger et al., “Not-so-heavy metal(s): Chemical Abundances in the Ultra-faint Dwarf Galaxies Eridanus IV and Centaurus I”, *ApJ*, vol. 1002, no. 2, id. 182, **2026**. [doi:10.3847/1538-4357/ae5a3e](https://doi.org/10.3847/1538-4357/ae5a3e)

Casey, Q. O., [and 21 others, including **M. Heiger**], “Deep Photometric Observations of Ultrafaint Milky Way Satellites Centaurus I and Eridanus IV”, *ApJ*, vol. 984, no. 2, id. 148, IOP, **2025**. [doi:10.3847/1538-4357/adc67e](https://doi.org/10.3847/1538-4357/adc67e).

M. Heiger et al., “Reading between the (Spectral) Lines: Magellan/IMACS Spectroscopy of the Ultrafaint Dwarf Galaxies Eridanus IV and Centaurus I”, *ApJ*, vol. 961, no. 2, IOP, **2024**. [doi:10.3847/1538-4357/ad0cf7](https://doi.org/10.3847/1538-4357/ad0cf7)

J. Bruce, T. S. Li, A. B. Pace, **M. Heiger**, Y.-Y. Song, and J. D. Simon, “Spectroscopic Analysis of Milky Way Outer Halo Satellites: Aquarius II and Boötes II”, *ApJ*, vol. 950, no. 2, Art. no. 167, IOP, **2023**. [doi:10.3847/1538-4357/acc943](https://doi.org/10.3847/1538-4357/acc943).

S. K. Sarbadhicary, **M. Heiger**, C. Badenes et al., “The RR Lyrae Delay-Time distribution: A novel perspective on models of old stellar populations.” *ApJ*, vol. 912, no. 2, IOP, **2021**. [doi:10.3847/1538-4357/abca86](https://doi.org/10.3847/1538-4357/abca86)

IN REVIEW

Aganze, C., [and 49 others, including **M. Heiger**] “The Cocytos Stream: A Disrupted Globular Cluster from our Last Major Merger?”, *arXiv:2504.11687*, **2025**. [doi:10.48550/arXiv.2504.11687](https://doi.org/10.48550/arXiv.2504.11687).

Observing Proposals

2025B **2 nights** Magellan/MIKE – *Col* (primary author)

The Most Metal-Poor Stream C-19: Probing Globular Cluster Formation and Enrichment with Detailed Chemical Abundances

2025A **3 half-nights** Magellan/MIKE – *Col*

Chemical Abundance Analysis of the Cocytos Stellar Stream

2024B **19.05 hours** Gemini/GHOST – *PI*

Perspectives on first stars and nucleosynthesis from a new candidate hyper metal-poor star

2024B **21 hours** VLT/UVES – *Col*

Messengers from the Early Universe: a pristine star discovered within the DESI footprint

2024A **29 hours** Gemini/GHOST – PI
Galactic Chemical Evolution at its Extreme using Detailed Abundances of Ultra-faint Dwarf Galaxy Stars

Awards, Fellowships, & Grants

- 2025 **Data Sciences Institute Doctoral Fellowship**, University of Toronto
- 2024 **Intl. Graduate Student Fellowship for Excellence in Doctoral Studies**, University of Toronto (cognate to the Ontario Graduate Scholarship)
- 2024 **Graduate Student Visitor Program**, One of two students selected to visit Gemini South based on meritorious application to program and successful observing proposal for thesis-related research
- 2021-2026 **Mary & Ron Martin International Graduate Student Fellowship**, University of Toronto
- 2021-2026 **C. A. Chant Fellowship**, University of Toronto
- 2020 **Honorable mention**, NSF Graduate Research Fellowship Program

Presentations

TALKS

- July 2026 **Near Field Cosmology in the Era of Big Data: Local Group and Beyond**, Fields institute
- July 2026 **2nd Annual Cosmic Horizons Conference**, University of Virginia
- April 2026 **STRAND seminar**, University of San Diego
- April 2026 **TASTY seminar**, University of Toronto
- March 2026 **AstroNuc 2026**, University of Arizona
- Dec. 2024 **Dark Energy Spectroscopic Instrument collaboration meeting**, Cancún, MX
- May 2024 **Canadian Astronomical Society | Société Canadienne d'Astronomie General Meeting**, Toronto, CA
- May 2024 **Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group**, University of Toronto
- Nov. 2023 **TASTY seminar**, University of Toronto

POSTERS

- Jan. 2026 **Data Sciences Institute Talent Showcase**, University of Toronto
- June 2025 **Galactic Frontiers II**, Dartmouth University
- July 2024 **Small Galaxies, Cosmic Questions II**, Durham University

Mentoring

* co-supervised

- 2025-2026 **Rosayla Coulthard**, University of Toronto; *Investigating the properties of carbon-enhanced metal-poor stars in Milky Way satellites with DESI*
- 2024-2026 **Isabelle Laing***, University of Toronto; *Measuring chemical abundances with data-driven machine learning*
- 2023 **Stephanie Lee***, University of Toronto; *Galactic archaeology with star clusters and streams*

Teaching Experience

- 2021-2026 **AST 101 & AST 201** – Teaching Assistant
12 semesters experience developing exams and assignments, leading tutorials and review sessions, running in-person observing and sidewalk astronomy outreach sessions for non-major introductory astronomy courses
- 2024 **AST 198** – Teaching Assistant
Developed course material including lectures and discussion questions, facilitated discussion groups, with focus on efficacy, accuracy, and clarity in science communication for first-year seminar course
- 2022-2024 **Directed Reading Program** Dept. of Physics – Supervisor
Lead year-long reading group for undergraduate students to develop critical reading skills and explore an area of scientific interest; topics included spectroscopy and chemical tagging (2022/23) and nuclear astrophysics (2023/24)

Outreach, Service, & Professional Experience

SERVICE AND LEADERSHIP

- 2024-2025 **Co-President**, Graduate Astronomy Students Association
- 2022-2024 **Secretary**, Graduate Astronomy Students Association
- 2021-2022 **Mediation Committee**, Graduate Astronomy Students Association
- 2023-2025 **Graduate Mentorship Program**, Mentor – Provide advice and support to first-year graduate students and ease the transition to a new environment and to the demands of graduate school
- 2022-2025 **Undergraduate Mentorship Program**, Mentor – Mentor undergraduate astronomy student to provide advice and support according to student's personal and career goals. Mentees are now in PhD programs at Aix-Marseille Université and the University of Arizona.

PROFESSIONAL EXPERIENCE

- 2021-2022 **Local Organizing Committee**, DESI Milky Way Survey Working Group Meeting, University of Toronto
- 2024-2026 **Local Organizing Committee**, Astrostatistics workshop STATSTRO, University of Toronto

OUTREACH

- 2025 **Astronomy on Tap**, Speaker – Public lecture to 400+ attendees at Toronto's largest astronomy event
- 2024 **AstroTours**, Speaker – Public lecture at monthly graduate-run outreach event
- 2022-2025 **AstroTours**, Volunteer